Lesson 3 Gears and Belt System

Objective: You will be able to:

-Explain the transmission of motion through gears and belts.

- Calculate mechanical advantage.

-use gear for daily life.

1-Gear

A gear is a wheel with teeth on its outer edge.



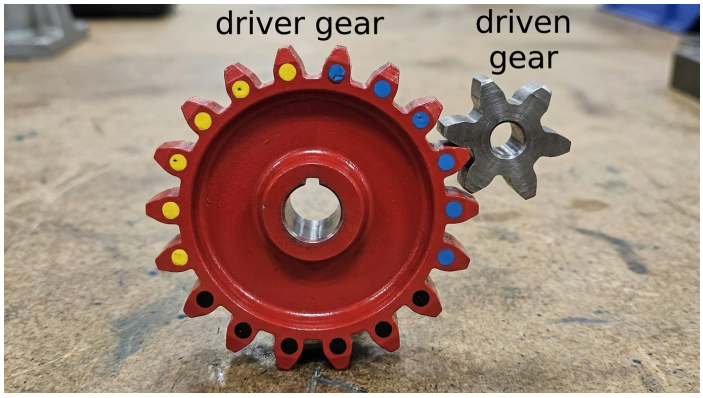
Gears are used to transmit turning force. They can also change the amount of force, speed and direction of rotation.

In the above image, the smaller gear is the ***driver*** or input gear. The driver's teeth cause the ***driven*** gear's teeth to rotate.

N_R, N_E the number of teeth on the driven and driver gear respectively.

This shows that if the driven gear has more teeth than the driver gear, then the gear train *amplifies* the driver moment.





And, if the driven gear has fewer teeth than the driver gear, then the gear system *reduces* the driver moment.

If the driven gear rotates more slowly than the driver gear, then the gear system is called a *speed reducer*. In this case, because the driven gear must have more teeth than the driver gear, the speed reducer will amplify the driver torque.

2.Mechanical advantage: $MA = \frac{F_R}{F_E}$

F_R, F_E are called load or resistance force and driving force respectively.

☞Relation between the number of turns and number of teeth:

$$\frac{N_R}{N_E} = \frac{t_E}{t_R} = \frac{d_E}{d_R}$$

N_R, N_E the number of teeth on the driven and driver gear respectively.

t_R, t_E the number of turns of the driven and driver gear respectively.

d_R, d_E the diameter of the driven and driver gear respectively.

Practice

1. A 36-tooth gear turns three times. It is connected to a 12-tooth gear. How many times does the 12-tooth gear turn?
2. A 12-tooth gear is turned two times. How many times will the 24-tooth gear to which it is connected turn?
3. A 60-tooth gear is connected to a 24-tooth gear. If the smaller gear turns ten times, how many turns does the larger gear make?
4. A 60-tooth gear is connected to a 72-tooth gear. If the smaller gear turns twelve times, how many turns does the larger gear make?
5. A 72-tooth gear is connected to a 12-tooth gear. If the large gear makes one complete turn, how many turns does the small gear make?

3- Belt System

A belt is a loop of flexible material used to mechanically link two or more rotating shafts. Suppose the belt motion has no slip.



R_R is the radius of driven wheel B. R_E is the radius of driver wheel A.

r_0 is the radius of rotational axis

T is the tension in the belt.

If the wheel A and B rotate at the same speed, the wheels are said to be in equilibrium.

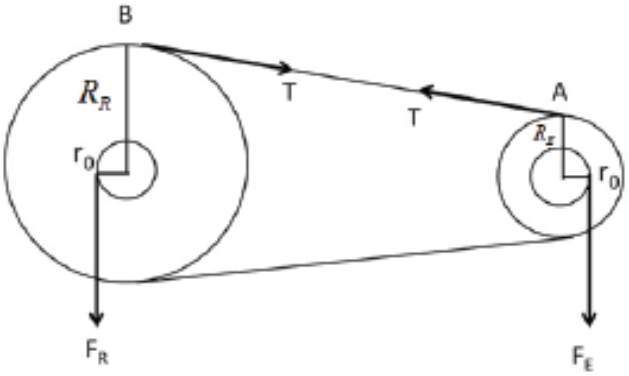
For wheel A: $T \times R_E - F_E \times r_0 = 0 \Rightarrow T = \dfrac{F_E \times r_0}{R_E}$ (1)

For wheel B: $F_R \times r_0 - T \times R_R = 0$ (2)

Substitute (1)in (2) $F_R \times r_0 - \dfrac{F_E \times r_0}{R_E} \times R_R = 0$

$$F_R = \frac{F_E}{R_E} \times R_R \Rightarrow \frac{F_R}{F_E} = \frac{R_R}{R_E} = MA$$

Mechanical Advantage: $\boxed{MA = \dfrac{F_R}{F_E} = \dfrac{R_R}{R_E}}$



☞Relationship between diameter and number of turns:

$$D_E.N_E = D_R.N_R$$

Where N_E, N_R number of turns of driver and driven wheel (r.p.m)

D_E, D_R diameter of driver and driven wheel

